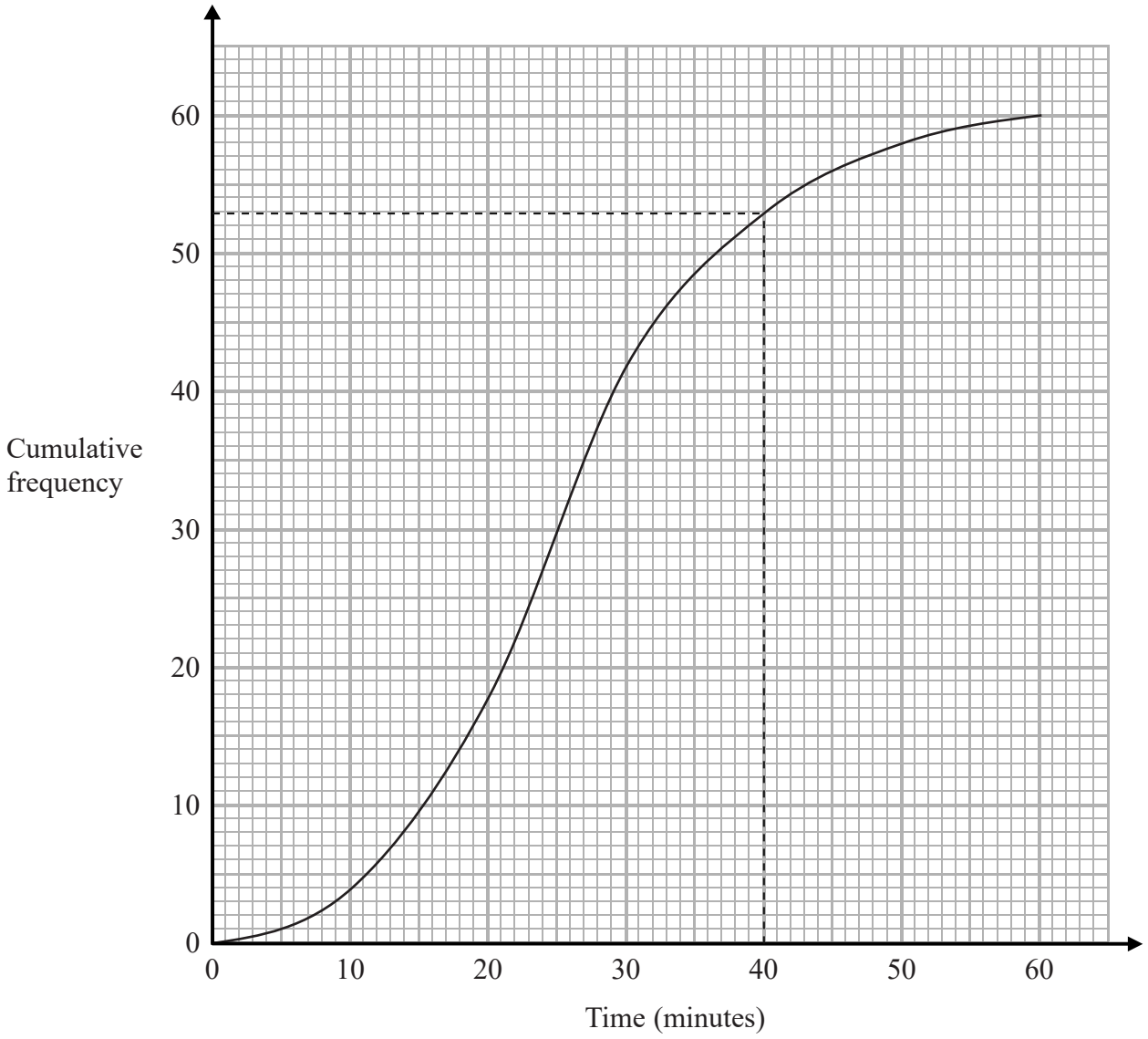
10 The cumulative frequency graph gives information about the number of minutes each of 60 people was in a shop.



(a) Find an estimate for the number of people who were in the shop for more than 40 minutes.

1 mark
Read from 40 min to the curve.

At 40 min: cumulative frequency ≈ 53

More than 40 min: 60 – 53 = 7 people

1 mark
Subtract from the total 60.

(2)

Stan has to use the graph to find an estimate for the lower quartile of the times.

Stan writes,

60 people were in the shop.
25% of 60 = 15
So the lower quartile of the times is 15 minutes.

(b) What mistake has Stan made?

15 is a cumulative frequency, not a time.

1 mark
Use the graph to find minutes.

Read across from CF 15 to the curve, then down to time.

(1)

(Total for Question 10 is 3 marks)